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Notations :

- Options shown in green color and with ✓ icon are correct.
- Options shown in red color and with ✗ icon are incorrect.

Question Paper Name :	PaperIIMathematics 10th Nov 2022Shift 1
Subject Name :	Paper II Mathematics
Creation Date :	2022-11-10 12:44:32
Duration :	150
Total Marks :	150
Display Marks:	Yes
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes
Calculator :	None
Magnifying Glass Required? :	No
Ruler Required? :	No
Eraser Required? :	No
Scratch Pad Required? :	No
Rough Sketch/Notepad Required? :	No
Protractor Required? :	No
Show Watermark on Console? :	Yes
Highlighter :	No
Auto Save on Console?	Yes
Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No

Show Reports :	No
Show Progress Bar :	No

Paper II Mathematics

Group Number :	1
Group Id :	1456343
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No

Mathematics

Section Id :	1456344
Section Number :	1
Section type :	Online
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	1456345
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 1 Question Id : 145634252 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The successor of 209999 is:

Options :

1. ✓ 210000

2. ✗ 209998

3. ✗ 201000

4. ✗ 219999

Question Number : 2 Question Id : 145634253 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Below given equation is under which property?

$$(13 \times 9) \times 4 = 13 \times (9 \times 4)$$

Options :

1. ✗ Closer property

2. ✗ Distribution property

3. ✗ Commutative property

4. ✓ Associative property

Question Number : 3 Question Id : 145634254 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is true?

Options :

1. ✖ The difference between the successor and the predecessor of a number is 1.
2. ✔ The difference between the successor and the predecessor of a number is 2.
3. ✖ All whole numbers are also natural numbers.
4. ✖ The difference between the successor of a number and the number itself is -1 .

Question Number : 4 Question Id : 145634255 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is true?

Options :

1. ✖ $(-6) + (-7) > (-6) - (-7)$
2. ✔ $(-6) + (-7) < (-6) - (-7)$
3. ✖ $(-6) + (-7) = (-6) - (-7)$
4. ✖ $((-6) + (-7)) \times (-1) = ((-6) - (-7))(-1)$

Question Number : 5 Question Id : 145634256 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The sum of two integers is -67 . If one of the integers is -43 , then the other integer is:

Options :

1. ✖ 24

2. ✖ 110

3. ✔ -24

4. ✖ -110

Question Number : 6 Question Id : 145634257 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The number 7.201201201201 ... is a/an:

Options :

1. ✖ Natural number

2. ✖ Integer

3. ✔ Rational number

4. ✖ Irrational number

Question Number : 7 Question Id : 145634258 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Express $3.\overline{323}$ as vulgar fraction:

Options :

1. ✔ $\frac{329}{99}$

2. ✖ $\frac{329}{999}$

3. ✖ $\frac{329}{990}$

4. ✖ $\frac{329}{9}$

Question Number : 8 Question Id : 145634259 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Find the value of x in the below equation:

$$\left(\frac{4}{5}\right)^2 \times \left(\frac{125}{64}\right)^{4x-2} = \left(\frac{25}{16}\right)^8$$

Options :

1. ✖ 0

2. ✖ 1

3. ✔ 2

4. ✖ 4

Question Number : 9 Question Id : 145634260 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The decimal expansion $\frac{63}{72 \times 175}$ is:

Options :

1. ✖ a non terminating decimal
2. ✖ an irrational number
3. ✔ a terminating decimal
4. ✖ an integer

Question Number : 10 Question Id : 145634261 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The value of $-5 + 34 + (-13) - 3 - 51 - (-4)$ is:

Options :

1. ✖ 34
2. ✖ -30
3. ✔ -34
4. ✖ 30

Question Number : 11 Question Id : 145634262 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The value of $529 - 823 - 61 - 1403 + 1635$ is:

Note: For this question, discrepancy is found in

question/answer. So, this question is ignored for all candidates.

Options :

1. 0
2. -1
3. 1
4. 2

Question Number : 12 Question Id : 145634263 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Simplify below equation:

$$\frac{1}{4} - \frac{2}{3} + \frac{5}{6} + \left(-\frac{1}{4}\right) - \frac{1}{12}$$

Options :

1. ✓ $\frac{1}{12}$
2. ✗ $\frac{17}{12}$
3. ✗ $-\frac{19}{12}$
4. ✗ $\frac{1}{6}$

Question Number : 13 Question Id : 145634264 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The value of $42 \div (9 - 7) \text{ of } 3 - 6 \times 3 + 10$ is:

Options :

1. ✖ 55

2. ✖ 0

3. ✖ 181

4. ✔ -1

Question Number : 14 Question Id : 145634265 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Simplify following equation:

$$\frac{6 + 6 \times 6}{3 \times 3 + 5} \div \frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} \div \frac{1}{2} \times \frac{1}{2}}$$

Options :

1. ✖ $\frac{1}{3}$

2. ✔ $\frac{3}{4}$

3. ✖ $\frac{4}{3}$

4. ✖ 3

Question Number : 15 Question Id : 145634266 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The value of $\sqrt{41 - \sqrt{21 + \sqrt{19 - \sqrt{9}}}}$ is:

Options :

1. ✖ 2

2. ✔ 6

3. ✖ 7

4. ✖ 5

Question Number : 16 Question Id : 145634267 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\sqrt{1 + \frac{27}{169}} = 1 + \frac{x}{13}$ then the value of x is:

Options :

1. ✔ 1

2. ✖ 2

3. ✖ 3

4. ✖ 4

Question Number : 17 Question Id : 145634268 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Convert $7.\overline{781}$ into vulgar fraction:

Options :

1. ✖ $7\frac{100}{990}$

2. ✖ $7\frac{784}{990}$

3. ✖ $7\frac{794}{990}$

4. ✔ $7\frac{43}{55}$

Question Number : 18 Question Id : 145634269 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The value of:

$$\frac{0.3 \times 0.3 \times 0.3 + 0.06 \times 0.06 \times 0.06}{0.6 \times 0.6 \times 0.6 + 0.12 \times 0.12 \times 0.12}$$

Options :

1. ✖ 0.27

2. ✖ 0.81

3. ✖ 0.25

4. ✔ 0.125

Question Number : 19 Question Id : 145634270 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If a man takes 36 minutes to cover a certain distance at speed of 5 kmph. Then the distance covered by him is:

Options :

1. ✖ 2.5 km

2. ✖ 2 km

3. ✔ 3 km

4. ✖ 3.5 km

Question Number : 20 Question Id : 145634271 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A goods train having a length of 350 meters passes through a tunnel of 1250 meters in 80 seconds. What is the speed of the train in km/hr?

Options :

1. ✖ 58 km/hr

2. ✓ 72 km/hr

3. ✗ 62 km/hr

4. ✗ 76 km/hr

Question Number : 21 Question Id : 145634272 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Sowjanya can do a job in 12 days and Ujwal in 18 days. Sowjanya, Ujwal and Kiran working together can do it in 6 days. In what time can Kiran do it alone?

Options :

1. ✗ 24 days

2. ✗ 20 days

3. ✗ 30 days

4. ✓ 36 days

Question Number : 22 Question Id : 145634273 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A pipe can fill a tank in 12 hrs. Due to leakage in the bottom of the tank it is filled in 16 hrs. How much time will the leak take to empty the completely filled tank?

Options :

1. ✖ 20 hrs

2. ✔ 48 hrs

3. ✖ 24 hrs

4. ✖ 36 hrs

Question Number : 23 Question Id : 145634274 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A can complete $\frac{2}{3}$ of a work in 5 days and B can complete $\frac{3}{5}$ of the work in 10 days. In how many days both A and B together can complete the work?

Options :

1. ✔ $5\frac{5}{29}$

2. ✖ $7\frac{5}{21}$

3. ✖ $8\frac{5}{6}$

4. ✖ $9\frac{5}{7}$

Question Number : 24 Question Id : 145634275 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A number is first increased by 50% and then it is further increased by 30%. The percentage increase in the original number is:

Options :

1. ✖ 40%

2. ✖ 80%

3. ✖ 50%

4. ✔ 95%

Question Number : 25 Question Id : 145634276 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the discount percentage on 'buy 5 get 3 free' offer?

Options :

1. ✖ 35.5%

2. ✖ 36.5%

3. ✖ 39.25%

4. ✔ 37.5%

Question Number : 26 Question Id : 145634277 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Arun purchased bag for ₹ 400 and sold it to Ujwal at 30% profit, Ujwal sold it to Dinesh with 25% loss and Dinesh sold it to Sathvika with 40% profit. The difference between the profits of Arun and Dinesh is:

Options :

1. ✖ ₹ 30

2. ✖ ₹ 34

3. ✔ ₹ 36

4. ✖ ₹ 38

Question Number : 27 Question Id : 145634278 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the annual installment Mr. Rajendhar should pay to clear his debt of ₹ 8505 which he borrowed from a private bank and return to in 3 years at the rate of interest 5% simple interest?

Options :

1. ✖ ₹ 2020

2. ✖ ₹ 2500

3. ✖ ₹ 2600

4. ✔ ₹ 2700

Question Number : 28 Question Id : 145634279 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Compound interest on ₹ 4000 for 9 months at rate of 40% p.a. compounded quarterly is:

Options :

1. ✖ ₹ 1424
2. ✖ ₹ 1224
3. ✔ ₹ 1324
4. ✖ ₹ 1524

Question Number : 29 Question Id : 145634280 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A dishonest merchant uses a false weight of 200 gm instead of 250 gm. What is his profit percent on selling?

Options :

1. ✖ 20%
2. ✖ 30%
3. ✖ 40%
4. ✔ 25%

Question Number : 30 Question Id : 145634281 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A shopkeeper bought 12 dozen eggs at the rate of ₹ 5 per egg. During transit 12 eggs were broken. He sold the remaining eggs at the rate of ₹ 6 per egg. What is his percentage of profit?

Options :

1. ✖ 5%
2. ✔ 10%
3. ✖ 20%
4. ✖ 15%

Question Number : 31 Question Id : 145634282 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Ravi, Prasad and Kiran enter into a partnership by investing ₹ 3000, ₹ 2000 and ₹ 5000 respectively. What are their respective shares in annual profit of ₹ 5600?

Options :

1. ✖ ₹ 1400, ₹ 1120, ₹ 2800
2. ✖ ₹ 1680, ₹ 1400, ₹ 1680
3. ✖ ₹ 1400, ₹ 1120, ₹ 1680
4. ✔ ₹ 1680, ₹ 1120, ₹ 2800

Question Number : 32 Question Id : 145634283 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

9 buckets are needed to fill a tank in 1 hour. How long will it take for only 6 buckets of the same type to fill that tank?

Options :

1. ✖ 72 min

2. ✖ 84 min

3. ✔ 90 min

4. ✖ 96 min

Question Number : 33 Question Id : 145634284 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

25 men can take complete a construction in 15 days working 8 hours a day. How many hours a day should 15 men work to complete the same construction in 25 days:

Options :

1. ✖ 10

2. ✔ 8

3. ✖ 12

4. ✖ 6

Question Number : 34 Question Id : 145634285 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

By the division algorithm, the value of $15 \bmod 11$ is?

Options :

1. ✖ 1

2. ✖ 5

3. ✖ 7

4. ✔ 4

Question Number : 35 Question Id : 145634286 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

By the division algorithm, the quotient when -1 is divided by 3 is:

Options :

1. ✖ 4

2. ✖ 2

3. ✔ -1

4. ✖ 0

Question Number : 36 Question Id : 145634287 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Which of the following are co-prime numbers?

Options :

1. ✖ 21 and 24

2. ✖ 39 and 63

3. ✖ 34 and 51

4. ✔ 14 and 15

Question Number : 37 Question Id : 145634288 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The sum of the prime factors of 105 is:

Options :

1. ✔ 15

2. ✖ 30

3. ✖ 51

4. ✖ 10

Question Number : 38 Question Id : 145634289 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $y + 1$ is a factor of $p + 3y + 3y^2 + y^3$, then the value of p is:

Options :

1. ✖ 0

2. ✖ -1

3. ✖ 2

4. ✔ 1

Question Number : 39 Question Id : 145634290 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

If $(x - 4)$ and $(x + 7)$ are the two factors of $x^2 + kx + q$, then the values of k and q respectively are:

Options :

1. ✖ 11, -28

2. ✖ -3, 21

3. ✖ -11, 21

4. ✔ 3, -28

Question Number : 40 Question Id : 145634291 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

If $6x^2 - x + 5(4m - 2)$ is divisible by $(x + 2)$ and the remainder is $2m - 2$, then the value of m is:

Options :

1. ✔ -1

2. ✖ 2

3. ✖ 3

4. ✖ -4

Question Number : 41 Question Id : 145634292 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Find the HCF of $2^3 \times 3^3 \times 5^2 \times 7^2$, $2^4 \times 3^2 \times 5^3 \times 7^2$ and $3 \times 5 \times 7 \times 11$.

Options :

1. ✖ 1155

2. ✖ 1260

3. ✔ 105

4. ✖ 360

Question Number : 42 Question Id : 145634293 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A man has two kinds of oils of volume 120 liters and 160 liters. He wants to sell them by filling in equal volume of tin.

What is the greatest volume of such tin?

Options :

1. ✖ 140 liters

2. ✔ 40 liters

3. ✖ 20 liters

4. ✖ 30 liters

Question Number : 43 Question Id : 145634294 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Three numbers are in the ratio 1 : 4 : 3 and their LCM is 1800. Then their HCF is:

Options :

1. ✖ 200

2. ✔ 150

3. ✖ 175

4. ✖ 120

Question Number : 44 Question Id : 145634295 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

For any $p, q (\neq 0)$ there exist unique integers x and y such that $p = qx + y, 0 \leq y < q$, if $q = 4$ then which is not value of y ?

Options :

1. ✖ 1

2. ✖ 2

3. ✖ 3

4. ✔ 4

Question Number : 45 Question Id : 145634296 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\log_{10}(2x - 1) + \log_{10} 2 = \log_{10}(x - 2) + 1$, then the value of x is:

Options :

1. ✖ 1

2. ✖ 2

3. ✖ 4

4. ✔ 3

Question Number : 46 Question Id : 145634297 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The value of $\log\left(\frac{3}{4}\right) + \log\left(\frac{4}{5}\right) + \log\left(\frac{5}{6}\right) + \log\left(\frac{6}{7}\right) + \log\left(\frac{7}{8}\right) + \log\left(\frac{8}{9}\right)$ is:

Options :

1. ✖ $-\log 1$

2. ✖ $\log 2$

3. ✔ $-\log 3$

4. ✖ $\log 9$

Question Number : 47 Question Id : 145634298 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The value of $\log 4 + \log 8 - \log 16$ is:

Options :

1. ✓ $\log 2$

2. ✗ $\log 4$

3. ✗ 1

4. ✗ $-\log 4$

Question Number : 48 Question Id : 145634299 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Express as a single logarithm:

$$(\log x - \log y) + 3\log z$$

Options :

1. ✗ $\log\left(\frac{xy}{3z}\right)$

2. ✗ $\log\left(\frac{x}{3yz}\right)$

3. ✗ $\log\left(\frac{xy}{z^3}\right)$

4. ✓ $\log\left(\frac{xz^3}{y}\right)$

Question Number : 49 Question Id : 145634300 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Simplify given equation:

$$2x^2 + 5x + 3 - 2(2x - 3) - (x^2 + x - 1)$$

Options :

1. ✗ $x^2 - 10x + 5$

2. ✗ $x^2 - 1x + 2$

3. ✗ $x^2 - 10x - 4$

4. ✓ $x^2 + 10$

Question Number : 50 Question Id : 145634301 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Express $x^2 - x - 2$ in factors form:

Options :

1. ✓ $(x + 1)(x - 2)$

2. ✗ $(x - 1)(x - 2)$

3. ✗ $(x + 1)(x + 2)$

4. ✖ $(x - 1)(x + 2)$

Question Number : 51 Question Id : 145634302 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Express $6x^2 - 5x - 91$ in factors form:

Options :

1. ✖ $(x - 13)(6x + 7)$

2. ✖ $(3x - 13)(2x - 7)$

3. ✖ $(6x - 13)(x + 7)$

4. ✔ $(3x - 13)(2x + 7)$

Question Number : 52 Question Id : 145634303 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the remainder if we divide $x^3 + 3x^2 - 4x + 5$ by $x - 2$?

Options :

1. ✖ 7

2. ✔ 17

3. ✖ 23

4. ✖ 5

Question Number : 53 Question Id : 145634304 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the value of k , if $x - 3$ is a factor of $2x^2 - 3x - (3k + 1)$?

Options :

1. ✔ $\frac{8}{3}$

2. ✖ $-\frac{1}{3}$

3. ✖ 3

4. ✖ $-\frac{3}{8}$

Question Number : 54 Question Id : 145634305 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $2x^3 - 3x^2 + 4kx - 5$ has a remainder -1 when divided by $x + 2$, then the value of k is:

Options :

1. ✖ -1

2. ✖ -2

3. ✖ -3

4. ✓ -4

Question Number : 55 Question Id : 145634306 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical
Correct Marks : 1 Wrong Marks : 0.33

The least number that is divisible by all the numbers from 3 to 6 (both inclusive) is:

Options :

1. ✗ 30

2. ✗ 40

3. ✗ 50

4. ✓ 60

Question Number : 56 Question Id : 145634307 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical
Correct Marks : 1 Wrong Marks : 0.33

The HCF of $\frac{15}{2}, \frac{6}{14}, \frac{7}{8}$ is:

Options :

1. ✓ $\frac{1}{56}$

2. ✗ $\frac{1}{54}$

3. ✗

$$\frac{21}{56}$$

4. ✖ $\frac{51}{56}$

Question Number : 57 Question Id : 145634308 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The roots of the equation $x^2 - x - 2 = 0$ are:

Options :

1. ✔ $-1, 2$

2. ✖ $1, 2$

3. ✖ $-1, -2$

4. ✖ $1, -2$

Question Number : 58 Question Id : 145634309 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Roots of the $2x^2 + 7x + 6 = 0$ are:

Options :

1. ✖ $\frac{3}{2}, -2$

2. ✖

$$-\frac{3}{2}, 2$$

3. ✖ $\frac{3}{2}, 2$

4. ✔ $-\frac{3}{2}, -2$

Question Number : 59 Question Id : 145634310 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The nature of the roots of $3x^2 - 6x + 5 = 0$ is:

Options :

1. ✖ One real root and other root is not a real root

2. ✖ Real and equal

3. ✖ Real and different

4. ✔ No real roots

Question Number : 60 Question Id : 145634311 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The sum of roots of the equation $5x^2 - 11x + 6 = 0$ is:

Options :

1. ✖ $\frac{6}{5}$

2. ✖ $-\frac{6}{5}$

3. ✖ $-\frac{11}{5}$

4. ✔ $\frac{11}{5}$

Question Number : 61 Question Id : 145634312 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $x = -2$ and $2x - 3y = 5$ then the value of y is:

Options :

1. ✖ 3

2. ✖ $\frac{1}{3}$

3. ✖ $-\frac{1}{3}$

4. ✔ -3

Question Number : 62 Question Id : 145634313 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $k + m = -3$ and $k - m = -7$ are two simultaneous linear equations, then the value of k and m respectively are:

Options :

1. ✖ 5, 2
2. ✔ -5, 2
3. ✖ 5, -2
4. ✖ -5, -2

Question Number : 63 Question Id : 145634314 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The pair equations $x = 2$ and $y = 3$ graphically represents:

Options :

1. ✔ Perpendicular lines
2. ✖ Parallel lines
3. ✖ Intersecting at origin
4. ✖ Intersecting at point $(-2, -3)$

Question Number : 64 Question Id : 145634315 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The graph of $Y = 5$ is a line parallel to the:

Options :

1. ✖ Y-axis
2. ✔ X-axis
3. ✖ both X and Y axis
4. ✖ passing through origin

Question Number : 65 Question Id : 145634316 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The straight line $2x - 5y = 10$ intersects Y-axis at:

Options :

1. ✖ $(2, 0)$
2. ✔ $(0, -2)$
3. ✖ $(5, 0)$
4. ✖ $(5, -2)$

Question Number : 66 Question Id : 145634317 Question Type : MCQ Option Shuffling : Yes Is
Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum
Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the solution of simultaneous equations, $4x - ky = 17$ and $x - 5y = 8$ is $(3, -1)$, then the value of k is:

Options :

1. ✖ 2

2. ✖ 3

3. ✖ 4

4. ✔ 5

Question Number : 67 Question Id : 145634318 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Rajini's sister age is 7 years more than her age. If product of their ages is 60, then what is the age of Rajini's sister age?

Options :

1. ✔ 12

2. ✖ 21

3. ✖ 14

4. ✖ 28

Question Number : 68 Question Id : 145634319 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

In a right-angle triangle, altitude is 7 cm less than its base. If the hypotenuse is 13 cm, then its altitude is:

Options :

1. ✔ 5 cm

2. ✖ 12 cm

3. ✖ 7 cm

4. ✖ 9 cm

Question Number : 69 Question Id : 145634320 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The ages of Shanthi and Keerthi are in the ratio of 7 : 3. Five years ago, product of their ages is 64. What is the age of Keerthi after 5 years?

Options :

1. ✔ 14

2. ✖ 8

3. ✖ 12

4. ✖ 18

Question Number : 70 Question Id : 145634321 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The difference of ages of Radha and Rama is 6 years and sum of their ages is 26. The product of their ages is:

Options :

1. ✖ 60

2. ✖ 120

3. ✖ 140

4. ✔ 160

Question Number : 71 Question Id : 145634322 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

A fraction becomes $\frac{1}{5}$ when 3 is subtracted from the numerator and is becomes $\frac{1}{3}$ when 5 is added to its denominator.

Sum of denominator and numerator is:

Note: For this question, discrepancy is found in question/answer. So, this question is ignored for all candidates.

Options :

1. 5

2. 4

3. 9

4. 12

Question Number : 72 Question Id : 145634323 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If A and B are sets, then $A \subset B$ means:

Options :

1. ✖ A subset of B
2. ✖ A belongs B
3. ✖ A improper subset of B
4. ✔ A proper subset of B

Question Number : 73 Question Id : 145634324 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

An empty set is _____ of every set.

Options :

1. ✔ subset
2. ✖ proper subset
3. ✖ power set
4. ✖ super set

Question Number : 74 Question Id : 145634325 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Reduce the rational expression $\frac{x^2-2x-8}{x^2-9x+20}$ to lowest terms:

Options :

1. ✖ $\frac{x-2}{x-5}$

2. ✔ $\frac{x+2}{x-5}$

3. ✖ $\frac{x+2}{x+5}$

4. ✖ $\frac{x-2}{x+5}$

Question Number : 75 Question Id : 145634326 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If A is set, then $(A')' =$

Options :

1. ✖ φ

2. ✖ A'

3. ✖ μ

4. ✔ A

Question Number : 76 Question Id : 145634327 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Sum of two numbers is -7 and their difference is 15 . The product of two numbers is:

Options :

1. ✓ -44

2. ✗ 25

3. ✗ 56

4. ✗ 22

Question Number : 77 Question Id : 145634328 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $A = \{0\}$, $B = \{3, 6, 7, 10\}$ and $C = \{4, 6, 8, 10\}$, then $(A \cup B) \cap C$ is:

Options :

1. ✗ $\{0\}$

2. ✗ $\emptyset$

3. ✓ $\{6, 10\}$

4. ✗ $\{6\}$

Question Number : 78 Question Id : 145634329 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If A and B are two sets, then $(A \cup B)^c =$

Options :

1. ✖ $A^c \cup B^c$

2. ✖ $\emptyset$

3. ✔ $A^c \cap B^c$

4. ✖ μ

Question Number : 79 Question Id : 145634330 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If A is non-empty set, then $A \cap \emptyset =$

Options :

1. ✖ A^c

2. ✖ 0

3. ✔ $\emptyset$

4. ✖ A

Question Number : 80 Question Id : 145634331 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If A is non-empty set, then $A \cap A^c =$

Options :

1. ✖ μ

2. ✔ $\emptyset$

3. ✖ A

4. ✖ A^c

Question Number : 81 Question Id : 145634332 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\sqrt{\frac{1-\cos \theta}{1-\sin \theta}}$, when $\theta = 30^\circ$?

Options :

1. ✖ 2

2. ✖ $-\sqrt{3}$

3. ✔ $\sqrt{2-\sqrt{3}}$

4. ✖ 0

Question Number : 82 Question Id : 145634333 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\frac{4 \cos \theta + 2 \sin \theta}{4 \sin \theta - 2 \cos \theta}$, when $\theta = 45^\circ$?

Options :

1. ✖ 0

2. ✖ 1

3. ✖ 2

4. ✔ 3

Question Number : 83 Question Id : 145634334 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\frac{\tan^2 60^\circ + \cot^2 30^\circ}{\tan^2 20^\circ - \sec^2 20^\circ}$?

Options :

1. ✔ -6

2. ✖ -2

3. ✖ 1

4. ✖ 0

Question Number : 84 Question Id : 145634335 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\frac{\tan 45^\circ}{\cot 45^\circ} + \frac{\sin 45^\circ}{\cos (-45^\circ)}$?

Options :

- 1. ✓ 2
- 2. ✗ 1
- 3. ✗ 0
- 4. ✗ -1

Question Number : 85 Question Id : 145634336 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\sec 33^\circ \operatorname{cosec} 57^\circ - \cot 57^\circ \tan 33^\circ$?

Options :

- 1. ✗ 0
- 2. ✓ 1
- 3. ✗ 2
- 4. ✗ -1

Question Number : 86 Question Id : 145634337 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\cot(A - B) = \sqrt{3}$ and $\cot(A + B) = \frac{1}{\sqrt{3}}$, $0 \leq A + B \leq 90^\circ$, then the value of $2A - B$ is:

Options :

1. ✖ 60°

2. ✖ 45°

3. ✔ 75°

4. ✖ 30°

Question Number : 87 Question Id : 145634338 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\frac{\cos^2 12^\circ + \cos^2 23^\circ}{2\sin^2 78^\circ + 2\sin^2 67^\circ}$?

Options :

1. ✖ 0

2. ✖ 1

3. ✔ $\frac{1}{2}$

4. ✖ 2

Question Number : 88 Question Id : 145634339 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\tan \theta = \frac{3}{4}$ then the value of $\frac{2\cos\theta - \sin\theta}{2\cos\theta + \sin\theta} = ?$

Options :

1. ✖ $\frac{7}{11}$

2. ✔ $\frac{5}{11}$

3. ✖ $\frac{8}{11}$

4. ✖ $\frac{5}{13}$

Question Number : 89 Question Id : 145634340 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $3 \cos \theta = 4 \sin \theta$ then the value of $\frac{2 \sin \theta - \sec^2 \theta + \cos \theta}{2 \sin \theta + \sec^2 \theta - \cos \theta} = ?$

Options :

1. ✖ $\frac{51}{15}$

2. ✖ $\frac{35}{15}$

3. ✖ $\frac{30}{157}$

4. ✔ $\frac{35}{157}$

Question Number : 90 Question Id : 145634341 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

From a certain point on the ground, the angle of elevation of the top of a pole is 30° . On moving 6 meters towards the pole, the angle of elevation becomes 60° . Then the height of the pole is:

Options :

1. ✓ $3\sqrt{3}$ m

2. ✗ $2\sqrt{3}$ m

3. ✗ $\sqrt{3}$ m

4. ✗ $\frac{2}{\sqrt{3}}$ m

Question Number : 91 Question Id : 145634342 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the coordinates of a point A are (x, y) . Distance of point A from $(1, 3)$ is equal to the distance of point A from point $(-3, 1)$. Then:

Options :

1. ✗ $2x = y$

2. ✓ $2x = -y$

3. ✗ $x = -2y$

4. ✗ $x = 2y$

Question Number : 92 Question Id : 145634343 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If a cable is bent to form a circle with radius 14 cm. If the cable re-bent to form a square, then its side will be:

Options :

1. ✖ 28 cm

2. ✔ 22 cm

3. ✖ 88 cm

4. ✖ 20 cm

Question Number : 93 Question Id : 145634344 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the perimeter of a rhombus is 100 cm and one of its diagonals is 20 cm. Then the area of rhombus is:

Options :

1. ✔ $100\sqrt{21} \text{ cm}^2$

2. ✖ $200\sqrt{21} \text{ cm}^2$

3. ✖ $50\sqrt{21} \text{ cm}^2$

4. ✖ $30\sqrt{21} \text{ cm}^2$

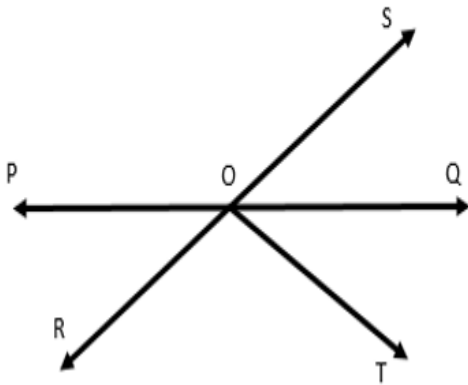
Question Number : 94 Question Id : 145634345 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If lines PQ and RS intersect at O and $\angle POR + \angle TOQ = 85^\circ$ and $\angle QOS = 30^\circ$ then the reflex of $\angle ROT = ?$



Options :

1. ✖ 85°

2. ✖ 95°

3. ✔ 265°

4. ✖ 195°

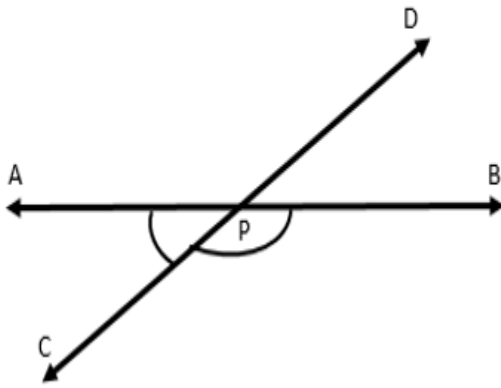
Question Number : 95 Question Id : 145634346 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If lines AB and CD intersect each other at a point P and $\angle APC : \angle CPB = 4 : 6$, then the $\angle DPA = ?$



Options :

1. ✓ 108°
2. ✗ 118°
3. ✗ 112°
4. ✗ 102°

Question Number : 96 Question Id : 145634347 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the transversal intersects two parallel lines AB and CD then which of the following true?

Options :

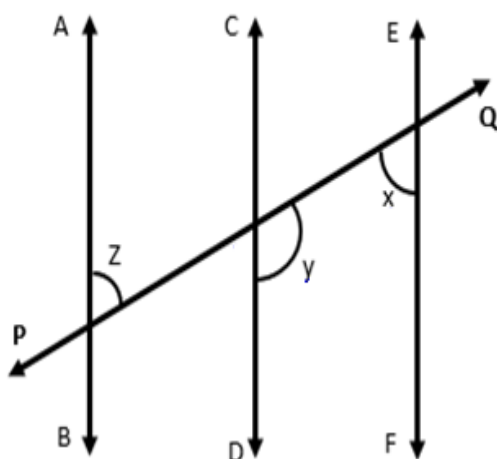
1. ✗ The pair of corresponding angles are different
2. ✓ The pair of corresponding angles are equal
3. ✓ The alternate interior and alternate exterior angles are always equal
4. ✗ The alternate interior and alternate exterior angles are always acute

Note: For this question, ambiguity is found in question/answer. Candidate will get full marks for this question if any of the correct options are chosen.

Question Number : 97 **Question Id :** 145634348 **Question Type :** MCQ **Option Shuffling :** Yes **Is Question Mandatory :** No **Calculator :** None **Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0 **Negative Marks Display Text :** 1/3 **Option Orientation :** Vertical

Correct Marks : 1 **Wrong Marks :** 0.33

In the following figure, if $AB \parallel CD$ and $CD \parallel EF$ and $y : z = 5 : 3$, then the value of x is :



Options :

1. ✖ 65.5°

2. ✔ 67.5°

3. ✖ 70.5°

4. ✖ 57.5°

Question Number : 98 **Question Id :** 145634349 **Question Type :** MCQ **Option Shuffling :** Yes **Is Question Mandatory :** No **Calculator :** None **Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0 **Negative Marks Display Text :** 1/3 **Option Orientation :** Vertical

Correct Marks : 1 **Wrong Marks :** 0.33

If an exterior angle of a triangle is 114° and its two interior opposite angles are in the ratio $1 : 2$ then the smallest angle between them is:

Options :

1. ✓ 38°

2. ✗ 57°

3. ✗ 37°

4. ✗ 36°

Question Number : 99 Question Id : 145634350 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The angles of the triangle are in the ratio $9 : 2 : 4$. The triangle is:

Options :

1. ✗ an isosceles triangle

2. ✗ an isosceles triangle

3. ✗ a right triangle

4. ✓ an obtuse-angled triangle

Question Number : 100 Question Id : 145634351 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\triangle ABC \cong \triangle PQR$ and three sides of triangle $\triangle PQR$ are $8\text{ cm}, 7\text{ cm}$ and 5 cm then what is the area of $\triangle ABC$?

Options :

1. ✖ 10

2. ✔ $10\sqrt{3}$

3. ✖ $5\sqrt{3}$

4. ✖ $20\sqrt{3}$

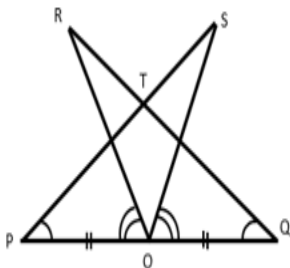
Question Number : 101 Question Id : 145634352 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If O is a midpoint of a line segment PQ and R and S are the two points as shown in the figure. Then which of the following is NOT correct?



Options :

1. ✖ $\angle POS = \angle QOR$

2. ✖ $\Delta SPO \cong \Delta RQO$

3. ✖ $PS = QR$

4. ✔ ΔPTQ is a right-angled triangle

Question Number : 102 Question Id : 145634353 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Sides of two similar triangles are in the ratio $8 : 6$. Then the areas of these triangles are in the ratio:

Options :

1. ✖ $8 : 6$

2. ✖ $16 : 12$

3. ✔ $16 : 9$

4. ✖ $12 : 9$

Question Number : 103 Question Id : 145634354 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\triangle ABC \sim \triangle PQR$ and $AB = 5$, $PQ = 8$, $QR = 10$ and $PR = 12$ then the perimeter of the triangle ABC is:

Options :

1. ✔ 18.75

2. ✖ 21.75

3. ✖ 30

4. ✖ 27

Question Number : 104 Question Id : 145634355 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If $\triangle ACB \sim \triangle PQR$, $\angle P = 40^\circ$, $\angle R = 65^\circ$, $PQ = 8$, $PR = 6$ and $AC = 6.5$ then which of the following is true?

Options :

1. ✖ $\angle C = 75^\circ$, $BC = 4.88$
2. ✖ $\angle B = 75^\circ$, $BC = 4.88$
3. ✖ $\angle B = 75^\circ$, $AB = 4.88$
4. ✔ $\angle C = 75^\circ$, $AB = 4.88$

Question Number : 105 Question Id : 145634356 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Which of the following statement is wrong?

Options :

1. ✖ All 3 medians divide the triangle into 6 smaller triangles of equal area.
2. ✖ All medians of each triangle intersect each other at single point.
3. ✖ Every triangle has 3 medians.
4. ✔ Each median of a triangle meets its opposite side in right angle.

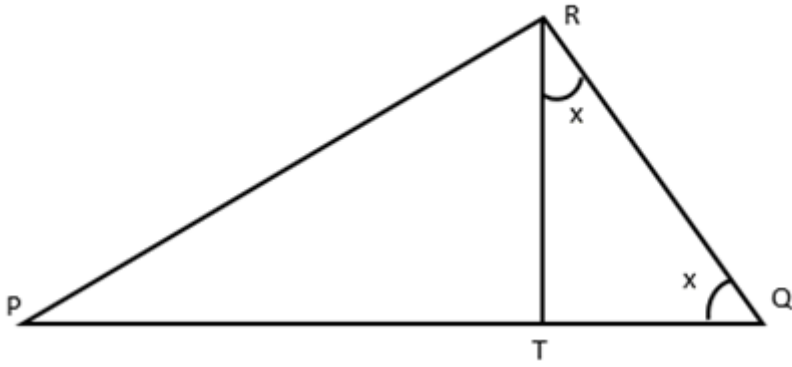
Question Number : 106 Question Id : 145634357 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If RT is an altitude in $\triangle PQR$, then what is the value of x ?



Options :

1. ✓ $x = 45^\circ$
2. ✗ $x = 30^\circ$
3. ✗ $x = 60^\circ$
4. ✗ $x = 50^\circ$

Question Number : 107 Question Id : 145634358 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

In a parallelogram PQRS, the angle R and S are both in the ratio $2 : 3$, then the angle P is?

Options :

1. ✗ 36°
2. ✓ 72°
3. ✗ 108°
4. ✗ 90°

Question Number : 108 Question Id : 145634359 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

A square of side 5 cm inscribed in a rectangle of length 10 cm and breadth 8 cm. What is the ratio of the area of the square to the remaining area of rectangle?

Options :

1. ✖ 11 : 16
2. ✔ 5 : 11
3. ✖ 5 : 16
4. ✖ 1 : 3

Question Number : 109 Question Id : 145634360 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The length of the rectangle is 7 cm more than its breadth. If the length of the diagonal is 13 cm, then the perimeter of rectangle is:

Options :

1. ✖ 48 cm
2. ✖ 17 cm
3. ✖ 12 cm
4. ✔ 34 cm

Question Number : 110 Question Id : 145634361 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

If T is the point on a side PQ of a rectangle $PQRS$. If $PS = \sqrt{13} \text{ cm}$, $ST = 7 \text{ cm}$ and $RT = 11 \text{ cm}$ then what is the length of PQ ?

Options :

1. ✓ $6 + 6\sqrt{3} \text{ cm}$
2. ✗ $6 + 12\sqrt{3} \text{ cm}$
3. ✗ $13 + 6\sqrt{3} \text{ cm}$
4. ✗ 12 cm

Question Number : 111 Question Id : 145634362 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

If $ABCD$ is a square, then which of the following is NOT true?

Options :

1. ✗ All the angles of square are right angled
2. ✗ All the sides are equal
3. ✓ Length of diagonal is double of the side
4. ✗ Diagonals are perpendicular to each other

Question Number : 112 Question Id : 145634363 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

If the area of a square is 72 cm^2 , then the sum of the length of its diagonal is:

Options :

1. ✓ 24 cm

2. ✗ 20 cm

3. ✗ 12 cm

4. ✗ 28 cm

Question Number : 113 Question Id : 145634364 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

If two tangents of a circle are parallel and 8 cm apart than its radius is:

Options :

1. ✗ 2 cm

2. ✓ 4 cm

3. ✗ 8 cm

4. ✗ 16 cm

Question Number : 114 Question Id : 145634365 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The locus of the point (x, y) where $x = a \cos \theta$ and $y = a \sin \theta$ is:

Options :

1. ✖ $x^2 + y^2 = 0$

2. ✖ $x^2 + y^2 = 1$

3. ✖ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

4. ✔ $x^2 + y^2 = a^2$

Question Number : 115 Question Id : 145634366 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Two-point P and Q are given, what is the locus of point A so that the $\angle PAQ$ is a right angle?

Options :

1. ✖ A Straight line

2. ✖ A Circle with Radius PQ

3. ✔ A Circle with Diameter PQ

4. ✖ A Square

Question Number : 116 Question Id : 145634367 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Find the diagonal of a square if its area is 48 m^2 .

Options :

1. ✖ $4\sqrt{2} \text{ m}$

2. ✖ $4\sqrt{3} \text{ m}$

3. ✖ $2\sqrt{6} \text{ m}$

4. ✔ $4\sqrt{6} \text{ m}$

Question Number : 117 Question Id : 145634368 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the side of square is decreased by 15%, what will be the percentage decreased in its area?

Options :

1. ✔ 27.75%

2. ✖ 30%

3. ✖ 32.25%

4. ✖ 2.75%

Question Number : 118 Question Id : 145634369 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The length of a rectangle is increased by 40%. By what percent would the width have to be decreased so as to maintain the same area?

Options :

1. ✖ $28\frac{2}{7}\%$

2. ✖ $28\frac{3}{7}\%$

3. ✔ $28\frac{4}{7}\%$

4. ✖ $28\frac{5}{7}\%$

Question Number : 119 Question Id : 145634370 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

An area of a land whose shape is parallelogram is 108 m^2 . Its base is thrice of its height. Then the height is:

Options :

1. ✖ 2 m

2. ✖ 4 m

3. ✔ 6 m

4. ✖ 8 m

Question Number : 120 Question Id : 145634371 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

An area of a land whose shape is parallelogram is 100 m^2 . Its height is 4 times of its base. Then the base is:

Options :

1. ✖ 2 m

2. ✖ 3 m

3. ✖ 4 m

4. ✔ 5 m

Question Number : 121 Question Id : 145634372 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A circular park has a path of uniform width around it. The difference between the outer and inner circumferences of the circular path is 88 m. Its width is:

Options :

1. ✔ 14 m

2. ✖ 24 m

3. ✖ 34 m

4. ✖ 44 m

Question Number : 122 Question Id : 145634373 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If a wire is bent into the shape of a square, then the area of the square is 324 cm^2 . When wire is bent into a semicircular shape, then the area of the semi-circle will be:

Options :

1. ✖ 14 cm^2
2. ✖ 77 cm^2
3. ✖ 154 cm^2
4. ✔ 308 cm^2

Question Number : 123 Question Id : 145634374 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The ratio of the outer and inner perimeters of a circular path is $5 : 3$. If the difference between their areas is $16\pi \text{ m}^2$, then inner diameter is:

Options :

1. ✖ 3 m
2. ✖ 4 m
3. ✔ 6 m
4. ✖ 8 m

Question Number : 124 Question Id : 145634375 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Find the side of maximum size of square which can be inscribed in a circle of radius 5 cm.

Options :

1. ✖ 5 cm

2. ✖ 10 cm

3. ✔ $5\sqrt{2}$ cm

4. ✖ $\frac{5}{\sqrt{2}}$ cm

Question Number : 125 Question Id : 145634376 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

What is the area of the largest circle that can be drawn inside a rectangle of sides 9 m × 7 m?

Options :

1. ✖ $\frac{7}{2}$ m²

2. ✖ $\frac{9}{4}\pi$ m²

3. ✖ $\frac{7}{2}\pi$ m²

4. ✔ $\frac{49}{4}\pi$ m²

Question Number : 126 Question Id : 145634377 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Volume of a cube is 64 cm^3 , then the surface area of the cube is:

Options :

1. ✖ 16 cm^2

2. ✖ 24 cm^2

3. ✔ 96 cm^2

4. ✖ 256 cm^2

Question Number : 127 Question Id : 145634378 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Two cubes each of volume 125 cm^3 are joined end to end, then the surface area of the resulting cuboid is:

Options :

1. ✖ 100 cm^2

2. ✖ 150 cm^2

3. ✖ 200 cm^2

4. ✔ 250 cm^2

Question Number : 128 Question Id : 145634379 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A conical vessel has radius 3 cm and its curved surface area is $15\pi \text{ cm}^2$. Then the height of the conical vessel is:

Options :

1. ✖ 8 cm

2. ✔ 4 cm

3. ✖ 2 cm

4. ✖ 1 cm

Question Number : 129 Question Id : 145634380 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The radii and height of two cylinders are in the ratio 5 : 3. The ratio of their surface areas is:

Options :

1. ✖ 5 : 3

2. ✔ 25 : 9

3. ✖ 3 : 5

4. ✖ 9 : 25

Question Number : 130 Question Id : 145634381 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the radii is 3 cm and surface area of the cylinder is $36\pi \text{ cm}^2$, then the height of the cylinder is:

Options :

1. ✖ 1 cm

2. ✖ 2 cm

3. ✔ 3 cm

4. ✖ 5 cm

Question Number : 131 Question Id : 145634382 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the radii is 8 cm and surface area of the cylinder is $272\pi \text{ cm}^2$, then the height of the cylinder is:

Options :

1. ✖ 5 cm

2. ✖ 6 cm

3. ✖ 8 cm

4. ✔ 9 cm

Question Number : 132 Question Id : 145634383 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

If the area of three adjacent faces of cuboid is 10 cm^2 , 80 cm^2 and 18 cm^2 respectively, then the volume of cuboid is:

Options :

1. ✖ 100 cm^3

2. ✖ 90 cm^3

3. ✔ 120 cm^3

4. ✖ 110 cm^3

Question Number : 133 Question Id : 145634384 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The base radii of two circular cones of the same height are in the ratio 2 : 5. The ratio of their volumes are:

Options :

1. ✖ 2 : 5

2. ✔ 4 : 25

3. ✖ 25 : 4

4. ✖ 5 : 2

Question Number : 134 Question Id : 145634385 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A conical hole is drilled in a circular cylinder of height 15 cm and base radius 6 cm. The height and base radius of the cone are also the same. The volume of the remaining cylinder is:

Options :

1. ✖ $36\pi \text{ cm}^3$
2. ✔ $360\pi \text{ cm}^3$
3. ✖ $72\pi \text{ cm}^3$
4. ✖ $720\pi \text{ cm}^3$

Question Number : 135 Question Id : 145634386 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A conical hole is drilled in a circular cylinder of height 9 cm and base radius 5 cm. The height and base radius of the cone are also the same. Then the ratio of volume of cone to volume of cylinder is:

Options :

1. ✖ 1 : 2
2. ✔ 1 : 3
3. ✖ 2 : 3
4. ✖ 3 : 5

Question Number : 136 Question Id : 145634387 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The volume of metallic sphere of diameter 10 cm is:

Options :

1. ✖ $\frac{100}{3}\pi \text{ cm}^3$

2. ✖ $\frac{200}{3}\pi \text{ cm}^3$

3. ✔ $\frac{500}{3}\pi \text{ cm}^3$

4. ✖ $\frac{1000}{3}\pi \text{ cm}^3$

Question Number : 137 Question Id : 145634388 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

The radii of two sphere are in the ratio 2 : 3. The ratio of their volumes is:

Options :

1. ✖ 2 : 3

2. ✖ 4 : 9

3. ✔ 8 : 27

4. ✖ 27 : 8

Question Number : 138 Question Id : 145634389 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Which statement is FALSE for the Primary Data Collection?

Options :

1. ✖ Primary data means first-hand information collected by an investigator.
2. ✖ It is collected for the first time.
3. ✔ For example, the address of a person taken from the telephone directory.
4. ✖ It is original and more reliable.

Question Number : 139 Question Id : 145634390 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Production in Maharashtra for different crops are given in the table:

Production in Tons/Month	Rice	Wheat	Jawar	Soyabean
2020	2.9	1.85	1.2	1.4
2019	3.28	2.3	1.4	1.6
2018	2.73	1.98	0.98	1.8
2017	3.11	2.7	1.1	2.3
2016	2.59	1.7	0.85	1.96

Find the difference between crop yields of highest and lowest production in the Maharashtra?

Options :

1. ✔ 9.08
2. ✖ 5.55
3. ✖ 4.08
4. ✖ 19.59

Question Number : 140 Question Id : 145634391 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

A quantity y varies inversely proportion to square of x if $y = 8$ when $x = 3$. What is the value of y when x is 6?

Options :

1. ✖ 1

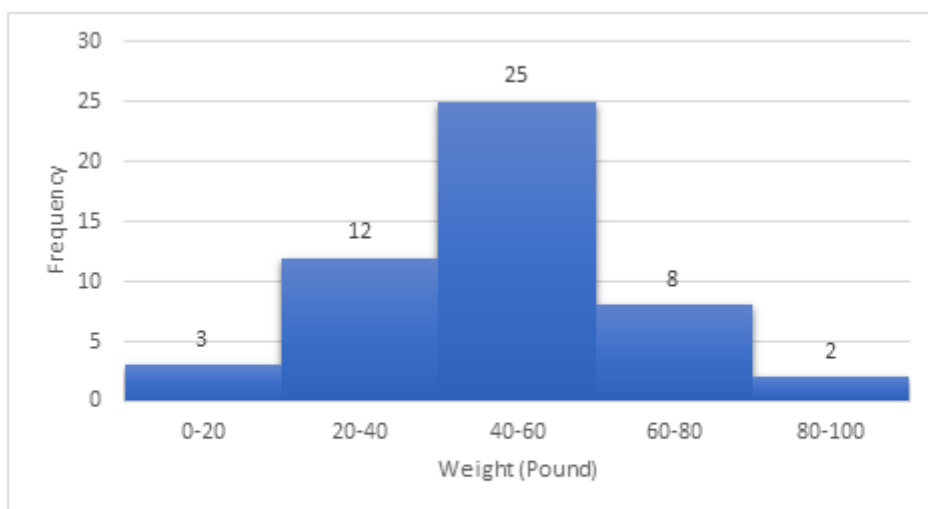
2. ✔ 2

3. ✖ 3

4. ✖ 4

Question Number : 141 Question Id : 145634392 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33



Find the mode for the data whose histogram is given above

Options :

1. ✖ 40

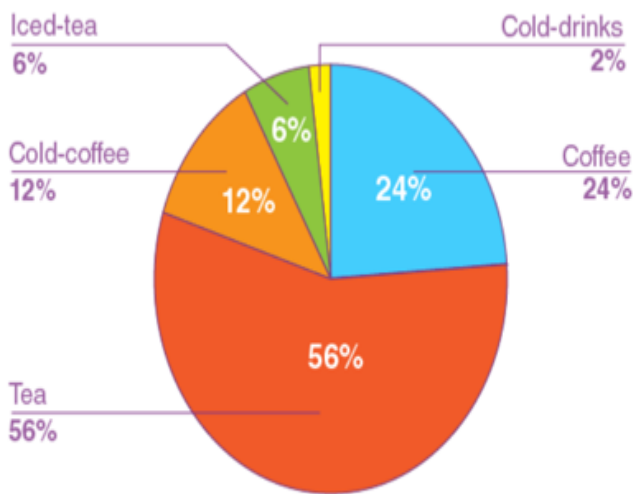
2. ✖ 45.67

3. ✔ 48.67

4. ✖ 60

Question Number : 142 Question Id : 145634393 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Pie chart based on a survey on 550 people about their favorite beverages is given as follows.



How many people who liked coffee and cold coffee?

Options :

1. ✖ 150

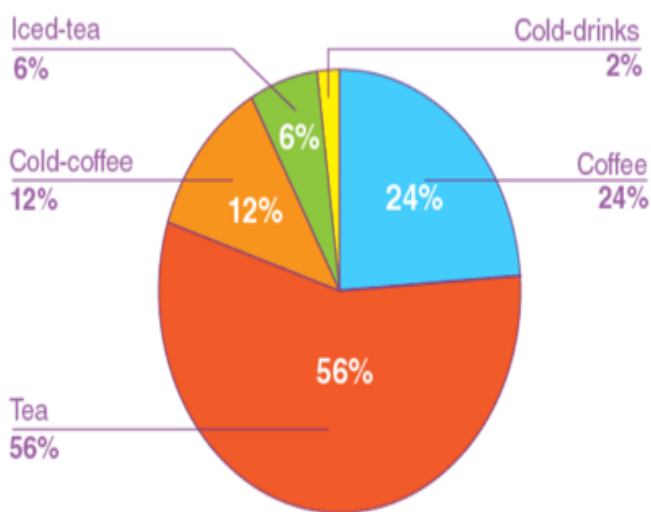
2. ✖ 176

3. ✔ 198

4. ✖ 206

Question Number : 143 Question Id : 145634394 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Pie chart based on a survey on a set people about their favorite beverages is given as follows.



If the number of people who liked tea is 140 then the total number of people is:

Options :

1. ✖ 200

2. ✖ 225

3. ✔ 250

4. ✖ 300

Question Number : 144 Question Id : 145634395 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Find mean of the following observations:

x	32	33	34	35	36
f	4	10	13	8	5

Options :

1. ✖ 30

2. ✖ 32

3. ✔ 34

4. ✖ 36

Question Number : 145 Question Id : 145634396 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Find the Mode for the data:

Class Interval	0–5	5–10	10–15	15–20	20–25
Frequency	5	3	7	2	6

Options :

1. ✖ 12

2. ✔ 12.22

3. ✖ 12.33

4. ✖ 13

Question Number : 146 Question Id : 145634397 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The height (in centimeters) of the members of a school football team have been listed below.

{142, 140, 130, 150, 160, 135, 158, 132}

Find the median of the above set.

Options :

1. ✖ 140

2. ✔ 141

3. ✖ 142

4. ✖ 150

Question Number : 147 Question Id : 145634398 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The age of the members of a weekend poker team has been listed below. Find the median of the above set.

{42, 40, 50, 60, 35, 58, 32}

Options :

1. ✔ 42

2. ✖ 40

3. ✖ 38

4. ✖ 36

Question Number : 148 Question Id : 145634399 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

Calculate the median for the following data:

Marks	0 - 20	20 - 40	40 - 60	60 - 80	80-100
Number of students	5	20	35	7	3

Options :

1. ✔ $\frac{320}{7}$

2. ✖ $\frac{270}{7}$

3. ✖ $\frac{230}{7}$

4. ✖ $\frac{200}{7}$

Question Number : 149 Question Id : 145634400 Question Type : MCQ Option Shuffling : Yes Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical Correct Marks : 1 Wrong Marks : 0.33

The scores obtained by two batsman A and B in 10 matches are given below.

Batsman A	30	44	66	62	60	34	80	46	20	38
Batsman B	34	46	70	38	55	48	60	34	45	30

The range of both are:

Options :

1. ✓ 60,40

2. ✗ 20,30

3. ✗ 80,70

4. ✗ 30,20

Question Number : 150 Question Id : 145634401 Question Type : MCQ Option Shuffling : Yes Is

Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum

Instruction Time : 0 Negative Marks Display Text : 1/3 Option Orientation : Vertical

Correct Marks : 1 Wrong Marks : 0.33

Consider marks of 10 students (out of 25) as 23, 10, 12, 23, 20, 12, 20, 15, 15, 25. The range is:

Options :

1. ✗ 10

2. ✗ 12

3. ✓ 15

4. ✗ 25